

Orientation Map

Read top to bottom = the pipeline in order. Glance, don't read aloud.

THE FLOW — 6 STEPS IN ORDER

1. `loadBrief`

Parse + Zod-validate the YAML/JSON brief.

| **WATCH:** needs min 2 products. `brandPalette` + `prohibitedWords` are optional.

2. `gatherAssets`

Reuse an existing asset, or generate one via Gemini.

| **WATCH:** not an LLM agent, just a for-loop. Throws on the first bad product = whole run dies.

3. `renderAspectRatios`

Sharp resizes each image to 1x1 / 9x16 / 16x9. Writes `base.png`.

| **WATCH:** fully sequential, no parallelism.

4. `applyOverlay`

SVG message bar (auto-shrinking font) + optional logo. Writes `final.png`.

| **WATCH:** the black bar it adds pollutes the color compliance check later.

5. `organizeOutputs`

Verify files, upload via storage adapter, write `manifest.json`.

| **WATCH:** upload failure is soft. Run still says "done", falls back to local path.

6. `checkCompliance`

Color score + logo vision check + prohibited-word scan. Writes `compliance.json`.

| **WATCH:** only step that isolates per-image failures. Know the 3 scorers cold.

THE PLUMBING

`routes/campaign.ts`

POST `/generate`, `/status`, `/manifest`, `/compliance`, image serving.

| **WATCH:** `/generate` is fire-and-forget. Returns `runId` instantly, run dies with the process.

lib/runStore.ts

In-memory Map of run state + cost ledger.

| **WATCH:** ephemeral, self-deletes after 1hr. Crash = total loss. Your weakest spot.

lib/paths.ts

Resolves OUTPUT_DIR.

| **WATCH:** ONE shared output dir, not per-run. Concurrent runs clobber each other.

lib/retry.ts

Exponential backoff + jitter.

| **WATCH:** only wraps Gemini + Dropbox. Claude prompt + vision are NOT retried.

workflow/mastra-compatible.ts

Hand-rolled Mastra-shaped runner.

| **WATCH:** API-shape only. Linear + in-memory. No resume/persistence.

lib/ai-clients.ts

Anthropic client + model constants.

| **WATCH:** claude-sonnet-4-5. No temperature set = non-determinism.

lib/storage/

Local + Dropbox live, S3/Azure scaffolded to throw.

| **WATCH:** Dropbox chosen for creative-team workflow, not tech preference.

lib/db/

Empty Drizzle stub, never imported.

| **WATCH:** this IS your "Postgres later." Honest deferral.

frontend/App.tsx

Polls /status every 750ms.

| **WATCH:** polling because Replit proxy buffers SSE. No SSE endpoint exists.

3 HOLES TO OWN FIRST

1. **Ephemeral run state** — a crash loses every in-flight run.
2. **Shared output dir** — concurrent runs overwrite each other's files.
3. **Thin compliance** — loose color threshold, vision reads text as a logo, words only scan your own message.